UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Parallel and multiprocessor architecture

a)
Give ONE (1) characteristic of Symmetric Multiprocessor (SMP).
b)
Discuss the meaning of "symmetric" in Symmetric Multiprocessor.
c)
Give ONE (1) advantage of Symmetric Multiprocessor organization.

Source: CSC429 (10).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Data representation and number systems

Convert 282810 to octal. Show your complete workflow.

Answer: 282810 = 54148

- *1 mark for identifying the target base.*
- *1 mark for a correct repeated-division or place-value method.*
- *1 mark for the correct digit sequence.*
- *1 mark for the correct base subscript.*
- *1 mark for a complete, consistent workflow.*

QUESTION A3 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Computer generations and history

a) Discuss the computer interface that is available during the third and fourth computer generations.
b) In your opinion, what is the leading cause that changed the computer interface in the third to the fourth computer generations?

Source: CSC429 (21).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Memory hierarchy and cache

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Describe FOUR (4) type of access methods based on hardware implementation of the storage device. Provide ONE (1) appropriate example for each access type.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: 38 - (-28)

a) Calculate 38 - (-28) using sign and magnitude.

Answer: left 38 -> 0 0100110; right -28 -> 1 0011100; 38 - (-28) = 66 -> sign-magnitude 0 1000010

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate 38 - (-28) using 1's and 2's complement.

Answer: left 38 -> 1's 00100110, 2's 00100110; right -28 -> 1's 11100011, 2's 11100100; 38 - (-28) = 66 -> 1's 01000010, 2's 01000010

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

QUESTION A7 (10 marks)

Topic: Input/output and interrupts

Question 8 is based on Figure 2.
interrupt
Interrupt
Handler B
H a n d l e r C
Interrupt
Handler A
Interrupt
Interrupt
Handler D Handler E
Figure 2: Multiple interrupts diagram
a) Describe maskable interrupt.
b) What is the sequence of the program execution if the defined priority is implemented in Figure 2?
c) What is the sequence of the program execution if the disable interrupt is implemented in Figure 2?
d) Which of the interruption method between defined priority or disable interrupt can finish Interrupt C earlier? Give your justification.

Source: CSC429 (14).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

Computer consists of four main components as illustrated in Figure 1. Each component has its own function in ensuring that the computer system works well. Question 4 is based on Figure 1.
Peripherals

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Central
Main
Processing
Memory
Unit
Computer
Systems
Interconnection
Input
Output
Communication
lines

Figure 1 : Components of computer

a) Explain THREE (3) types of system interconnection.

b) State TWO (2) sub-components of central processing unit and its function.

Source: CSC429 (3).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

a) Given the size of an address bus is 16 bit and the size of the data bus is 64 bit.

i) Determine the size of data block. Give appropriate justification for such answer.

ii) What is the usage of address bus?

iii) What happens if a wider address bus is set?

b) Discuss TWO (2) types of bus timing.

Source: CSC429 (8).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Input/output and interrupts

a)

Discuss TWO (2) functions of an input/output module.

b)

External input/output devices can be broadly categorized into three categories. List TWO (2) of the categories and provide an appropriate example of devices for each categories.

c)

Name ONE (1) input/output technique that requires active CPU intervention and ONE (1) technique that does not require CPU intervention.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 1 and Table 1.
z=A2-B*C+jl_
D+5
Figure 1
write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 1 using instructions for zero-address, one-address, and three-address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.
Table 1: Machine instruction
Three-address
One-address
Zero-address
ADD (p,q,r)
LOAD p
PUSH p
suB (p,q,r)
STOR p
POP p
MUL (p,q,r)
ADD p
ADD
DIV (p,q,r)
SUB p
SUB
MUL p
MUL
DIV p
DIV

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END OF MARKING SCHEME

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